■ TOP SECRET // CONTROLLED FORENSIC SPECIFICATION // NOFORN ■

PROJECT CYGNUS: QUANTUM-SAFE FORENSIC WATERMARKING SPECIFICATION

DOCUMENT ID:	CYGNUS-2026-ALPHA-09
SECURITY CLEARANCE:	LEVEL 5 (POST-QUANTUM CELL)
CRYPTOGRAPHIC SUITE:	ML-KEM-768 & ML-DSA-65 (NIST FIPS 203/204)
DISPOSITION:	FORENSIC VERIFICATION REFERENCE DATASET

1. OPERATIONAL OBJECTIVE

This controlled document serves as the benchmark forensic test payload for the Quantum-Safe Forensic Watermarking System (QSFW). It is distributed under an air-gapped broadcast model where recipient decryption dynamically embeds a session-unique, imperceptible forensic watermark bound to post-quantum digital signatures.

2. FORENSIC ATTRIBUTION GUARANTEE

Each authorized recipient decrypting this document produces an independent tamper-evident block committed to the distributed offline ledger. If this document is exfiltrated, photographed, or copied, the extraction pipeline will isolate the watermark pattern and cross-reference the canonical ledger records to establish non-repudiable legal attribution.

3. CRYPTOGRAPHIC VERIFICATION PROOFS

Integrity can be independently verified via the Cryptographic Verification Center using Document Hash validation, ML-DSA post-quantum signature verification, Merkle tree inclusion proofs, and ledger chain cross-validation.

■ END OF CLASSIFIED CYGNUS SPECIFICATION ■